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Lab 3: Architectural Justification — Internal Micro service Catalog & Health Portal

Architecture Selection: We chose Micro services Architecture for the Internal Micro service Catalog & Health Portal System.

Architectural Choice:

We selected Micro services Architecture because the system itself is designed to monitor and manage micro services, making it the most natural and fitting choice.

Two Reasons:

1. Each component (Service Catalog, Health Monitor, Alert Manager) can be independently deployed and scaled. If health monitoring receives heavy traffic, it can scale without affecting the catalog service.
2. Fault isolation ensures that if the Alert Manager fails, the Service Catalog and Health Monitor continue functioning independently, maintaining overall system availability.

Security Advantage:

The API Gateway acts as a single entry point, enforcing authentication and authorization before any request reaches internal components, reducing the attack surface.

Performance Benefit:

Each micro service component communicates asynchronously, allowing the Health Monitor to continuously poll services without blocking other operations, improving overall system responsiveness.